

**Ablation Study: Cross-Validated Confusion Matrices**  
(Target: High-Mobility p-type Semiconductors,  $m_p < 1.0 m_e$ )

**1. Full Pipeline**  
(F2=0.608)

| Actual Class | Predicted Class |           |
|--------------|-----------------|-----------|
|              | Low-Mob.        | High-Mob. |
| Low-Mob.     | 7806            | 785       |
| High-Mob.    | 90              | 355       |

**2. Exclude SMOTE**  
(F2=0.449)

| Actual Class | Predicted Class |           |
|--------------|-----------------|-----------|
|              | Low-Mob.        | High-Mob. |
| Low-Mob.     | 8515            | 76        |
| High-Mob.    | 262             | 183       |

**3. Exclude Feat. Select**  
(F2=0.572)

| Actual Class | Predicted Class |           |
|--------------|-----------------|-----------|
|              | Low-Mob.        | High-Mob. |
| Low-Mob.     | 7752            | 839       |
| High-Mob.    | 107             | 338       |

**4. Exclude XGBoost**  
(F2=0.579)

| Actual Class | Predicted Class |           |
|--------------|-----------------|-----------|
|              | Low-Mob.        | High-Mob. |
| Low-Mob.     | 7621            | 970       |
| High-Mob.    | 85              | 360       |

**5. Dummy Baseline**  
(F2=0.000)

| Actual Class | Predicted Class |           |
|--------------|-----------------|-----------|
|              | Low-Mob.        | High-Mob. |
| Low-Mob.     | 8591            | 0         |
| High-Mob.    | 445             | 0         |

**6. RUS Pipeline**  
(F2=0.550)

| Actual Class | Predicted Class |           |
|--------------|-----------------|-----------|
|              | Low-Mob.        | High-Mob. |
| Low-Mob.     | 7246            | 1345      |
| High-Mob.    | 59              | 386       |